

Aspiring bioinformatician with a strong background in molecular biology and immunology, seeking to deepen expertise in multi-omics, data integration, and translational cancer research. Motivated to contribute to high-impact scientific projects and pursue a PhD in systems biology or immuno-oncology.

SKILLS

Bioinformatics	Transcriptomics (bulk and spatial), Seurat, Harmony, Nextflow, R, Python, Multi-omics Integration
Data Science	Dimensionality Reduction, Clustering, Differential Expression, Feature Selection, Statistical Modeling
Technical Tools	Git, Anaconda, LaTeX, Shell scripting, High Performance Computing (HPC)
Languages	French (native), English (fluent), Spanish and Italian (notions)

EXPERIENCE

Master 2 Bioinformatics Intern – Spatial Transcriptomic Analysis
CRCL – Centre de Recherche en Cancérologie de Lyon

February 2025 — Present
Lyon, France

- Analyzed spatial transcriptomics data (10X Visium) from 15 triple-negative breast cancer (MpBC) patients.
- Integrated expert-annotated phenotypic cell types to characterize tumor heterogeneity.
- Developed and optimized normalization, feature selection, and batch effect correction pipelines (Seurat, Harmony).
- Identified spatially enriched markers and distinct transcriptional profiles among tumor subtypes.
- Contributed to ongoing research on rare breast cancer evolution and potential PhD project.

Master 1 Bioinformatics Intern – Multi-omics Analysis
CLB – Centre Léon Bérard

April 2024 — August 2024
Lyon, France

- Performed integrative analysis of genomic, transcriptomic, and proteomic data from the Profiler cancer cohort.
- Conducted differential expression and pathway enrichment to identify molecular signatures associated with clinical subtypes.
- Implemented reproducible pipelines in R and Nextflow for data preprocessing and normalization.
- Applied MOFA to identify latent factors capturing shared and specific sources of variation across omics layers.

Master 2 Immunology Research Intern – Immunopathology and Immuno-oncology
CRCL – Centre de Recherche en Cancérologie de Lyon

January 2023 — June 2023
Lyon, France

- Investigated the regulation of NF-κB signaling pathway in response to pro-inflammatory stimuli in human cell lines.
- Performed cell culture, Western blotting, immunofluorescence, and qRT-PCR to assess pathway activation and gene expression.
- Contributed to functional assays exploring the role of NF-κB in cellular stress and apoptosis.
- Analyzed experimental data to characterize molecular mechanisms of inflammation-related gene regulation.

Master 1 Molecular Biology Research Intern – RNA Delivery
CIRI – Centre International de Recherche en Infectiologie

April 2022 — July 2022
Lyon, France

- Developed and characterized a novel RNA delivery system based on the endogenous retroviral protein hPEG10.
- Engineered non-viral particles for mRNA delivery into primary human cells.
- Performed molecular biology techniques including RT-PCR, qPCR, enzymatic digestions, and Western blotting.
- Contributed to the production and analysis of pseudo-viral particles and maintained primary cell cultures.
- Collaborated within a research team and presented results through scientific communication.

EDUCATION

Spring School on AI & Machine Learning in Biology and Health, AI4BioMed, UCBL1

April 7-9, 2025

Bioinformatics Master's Degree, Claude Bernard Lyon 1 University — (UCBL1)

September 2023 — Present

- Master 2: All courses in Session 1 completed (grade and rank not yet available).
 - UE : "Génomique En Écologie Et Évolution 1", Grade : 12.25/20, Rank : 34/89
 - UE : "Modélisation Des Réseaux Métaboliques", Grade : 12.75/20, Rank : 9/15
 - UE : "Visualisation De Données Biologiques", Grade : 16/20, Rank : 3/21
 - UE : "Programmation Web Pour La Bioinformatique", Grade : 15/20, Rank : 10/18
 - UE : "Statistiques Bayésiennes Et Applications", Grade : 16/20, Rank : 1/17
 - UE : "Algorithmique Avancée Pour La Bio-Informatique", Grade : 13.20/20, Rank : 6/13

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Lyon, France

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Jordan Dutel

Bioinformatics Engineer – Junior Level

Portfolio: github.com/jdutel42

linkedin.com/in/jordan-dutel

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- UE : "Projet En Bioinformatique 3", Grade : 14.66, Rank : 15/22
 - UE : "Techniques D'Apprentissage Automatique", Grade : 14.40, Rank : 14/44
 - UE : "Analyse De Données", Grade : 14.75, Rank : 8/33
 - UE : "Anglais Pour La Communication Professionnelle", Grade : 14.33, Rank : 16/28
 - Master 1:
 - Session 1: Grade = 13.213/20, Rank: 12/22.
 - Session 2: Grade = 13.816/20 (no rank available).

Molecular and Cellular Biology Master's Degree, *Claude Bernard Lyon 1 University — (UCBL1)*
(Specialization in Immunology, Immunopathology, & Immunotherapy)

September 2021 — July 2023

- Master 2:
 - Session 1: Grade = 13.225/20, Rank: 10/12.
- Master 1:
 - Session 1: Grade = 13.116/20, Rank: 46/75.

Genetics Bachelor's Degree, *Claude Bernard Lyon 1 University — (UCBL1)*

September 2018 — June 2021

Licence 3:

- Session 1: Grade = 11.763/20, Rank: 39/129.
- Session 2: Grade = 12.509/20, Rank: 6/76.

PACES (Première Année Commune aux Études de Santé), *Jean Monnet University*

September 2016 — June 2018

Level: "Reçus-collés"

Scientific Baccalauréat (Honors: 'Assez Bien'), *Lycée L'Astrée, France*

June 2016